Laboratory 1a: Rule-Based Security Detection and MITRE ATT&CK

Course: AI-Driven Cybersecurity
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1. Initial Findings

The initial dashboard displayed two findings:

- **T1110 — Brute Force:** Three failed password attempts associated with source IP `172.18.0.5`.
- **T1046 — Network Service Discovery:** A network-scan log entry associated with source IP `172.18.0.7`.

2. Simulator Results

During the simulation, new source IPs appeared, including `172.18.0.22` and `172.18.0.33`. Multiple failed-password events from `172.18.0.22` triggered T1110 findings, while an explicit network-scan message from `172.18.0.33` triggered a T1046 finding.

3. Key Observation

The total finding count increased as the simulator generated additional events. Unlike controlled experiments, the simulation was less reproducible because events occurred over time. The findings indicate patterns matching detection rules but do not independently confirm malicious activity or a successful compromise.

4. SOC Narrative

The initial dashboard displayed two findings: a T1110 (Brute Force) finding associated with source IP 172.18.0.5 and a T1046 (Network Service Discovery) finding associated with 172.18.0.7. During the simulator activity, additional source IPs appeared, including 172.18.0.22 and 172.18.0.33. Multiple failed-password log entries from 172.18.0.22 triggered T1110 findings, while an explicit network-scan message from 172.18.0.33 triggered a T1046 finding. The total finding count increased as the simulator continued generating events. These results demonstrate how rule-based detection identifies log patterns associated with known MITRE ATT&CK techniques. However, the findings alone do not confirm malicious intent, a successful compromise, or actual network scanning. Further investigation and corroborating evidence would be required to determine whether the observed activity represents a genuine security incident.